



Caatinga Bank

An aerial photograph of a rugged limestone karst landscape. The terrain is characterized by numerous sharp, jagged rock peaks and ridges, some of which are covered in dense green vegetation. The overall scene is a mix of grey rock and green shrubs, creating a textured and dramatic landscape. The word "MONGABAY" is written in a white, serif font inside a thin white rectangular border in the upper right corner.

MONGABAY



1.4M
Family Farmers

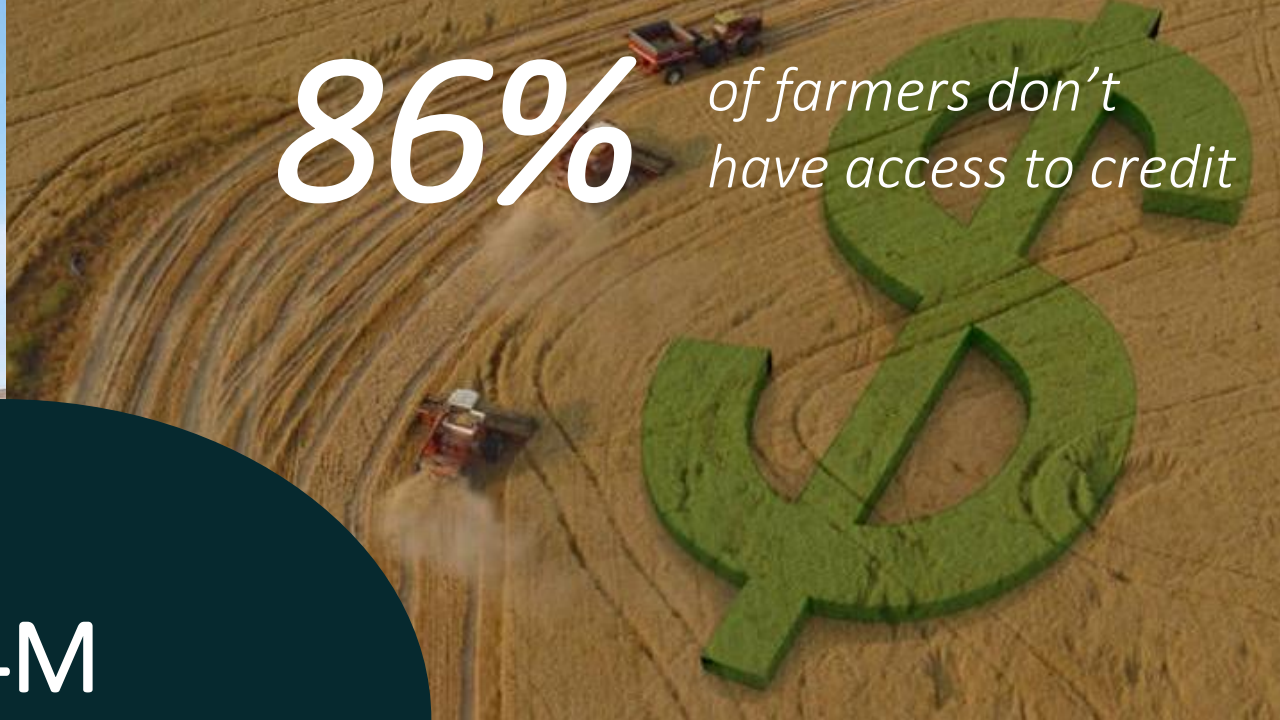


85% *of land is under
desertification*

**1.4M
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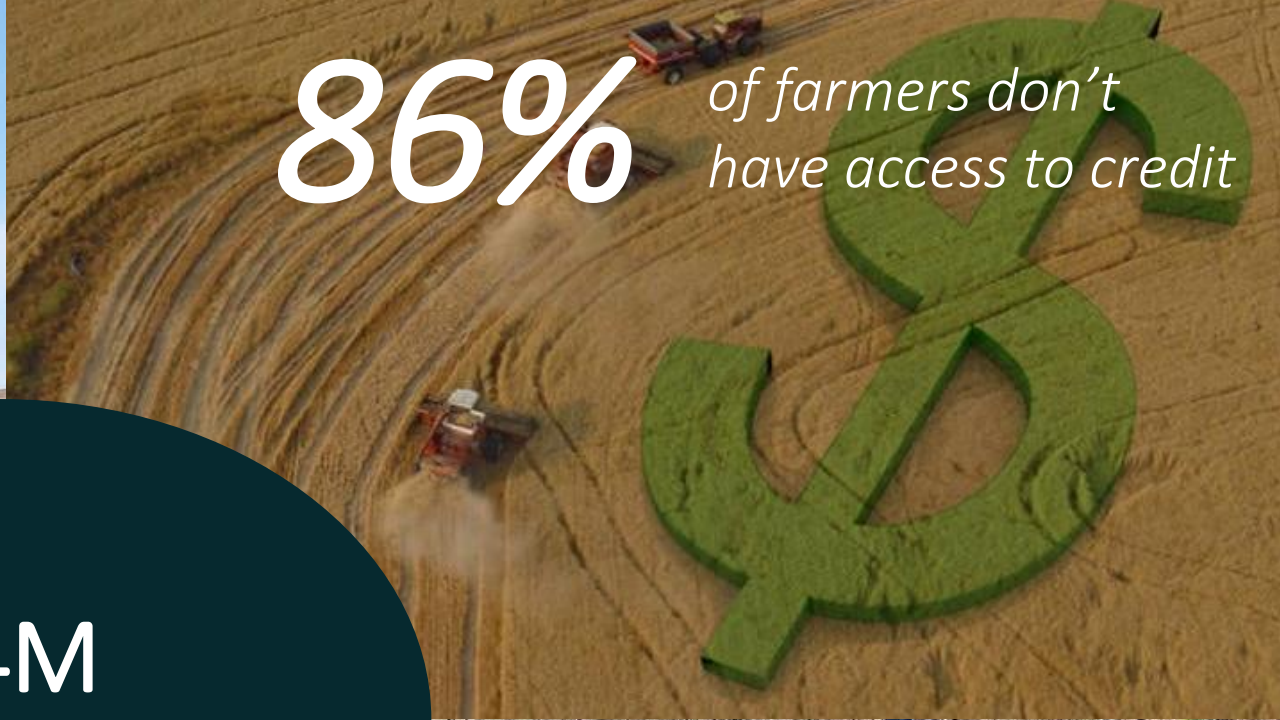


86% *of farmers don't
have access to credit*

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U\$116 income/hectare



85% of land is under desertification



86% of farmers don't have access to credit

1.4M
Family Farmers



21% of the population suffers from starvation



U\$116 income/hectare

Caatinga Bank provides working capital loans and technical assistance for family farmers while securing stable futures contracts



WORKING CAPITAL & ASSETS

Loan with the **money** needed to run the farm operation along with **seeds** and compost



TECHINICAL ASSISTANCE

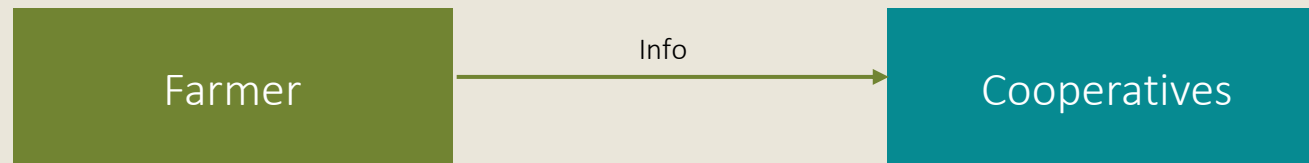
Partner with local companies to provide knowledge about **sustainable practices** and technical knowledge



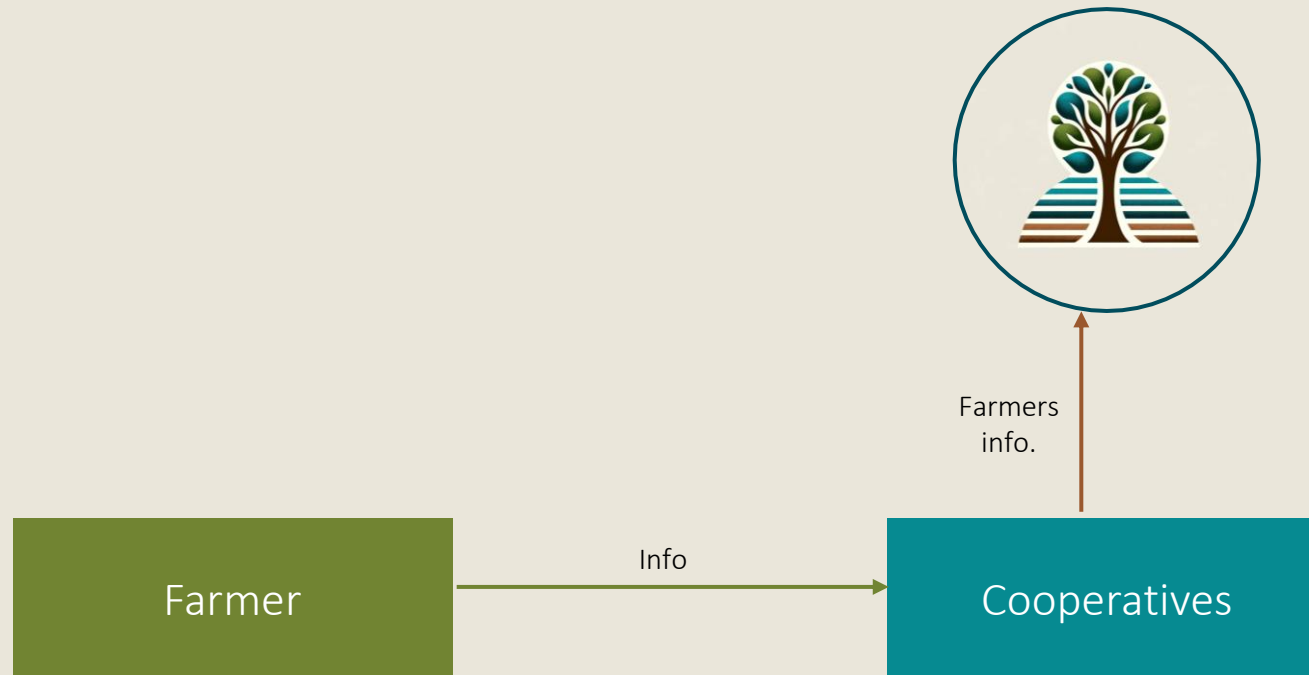
FUTURE CONTRACTS

Secure sales of farmers production to decrease risk of default and predictable income for farmers

Caatinga Bank will contact farmers through cooperatives to achieve higher capillarity

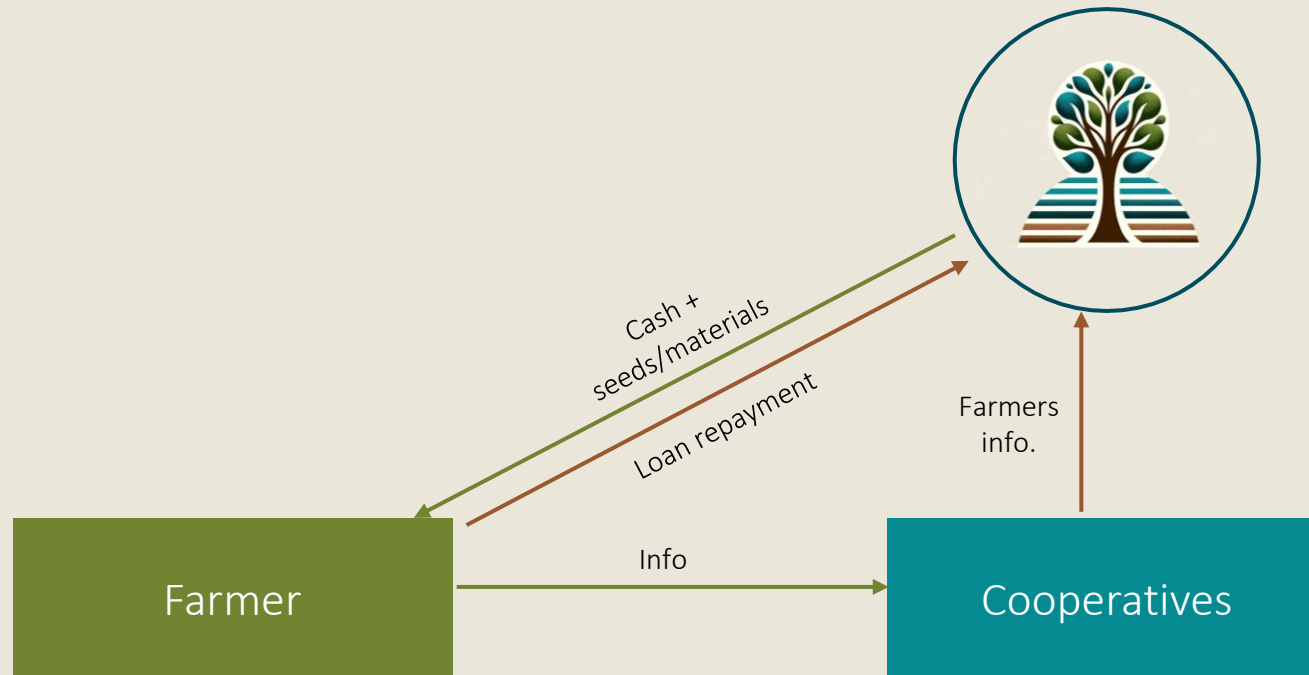


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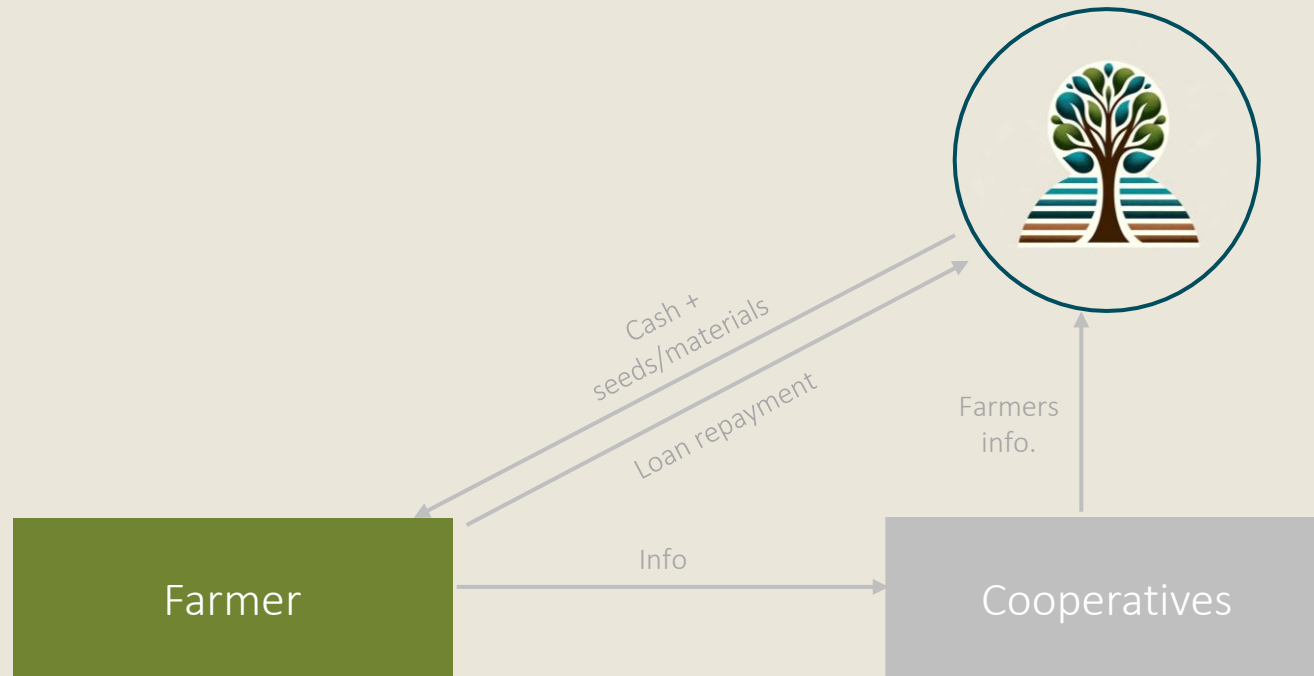
- Criteria to approve for credit:
1. Referral from a cooperative
 2. Personal Documents
 3. Size of land

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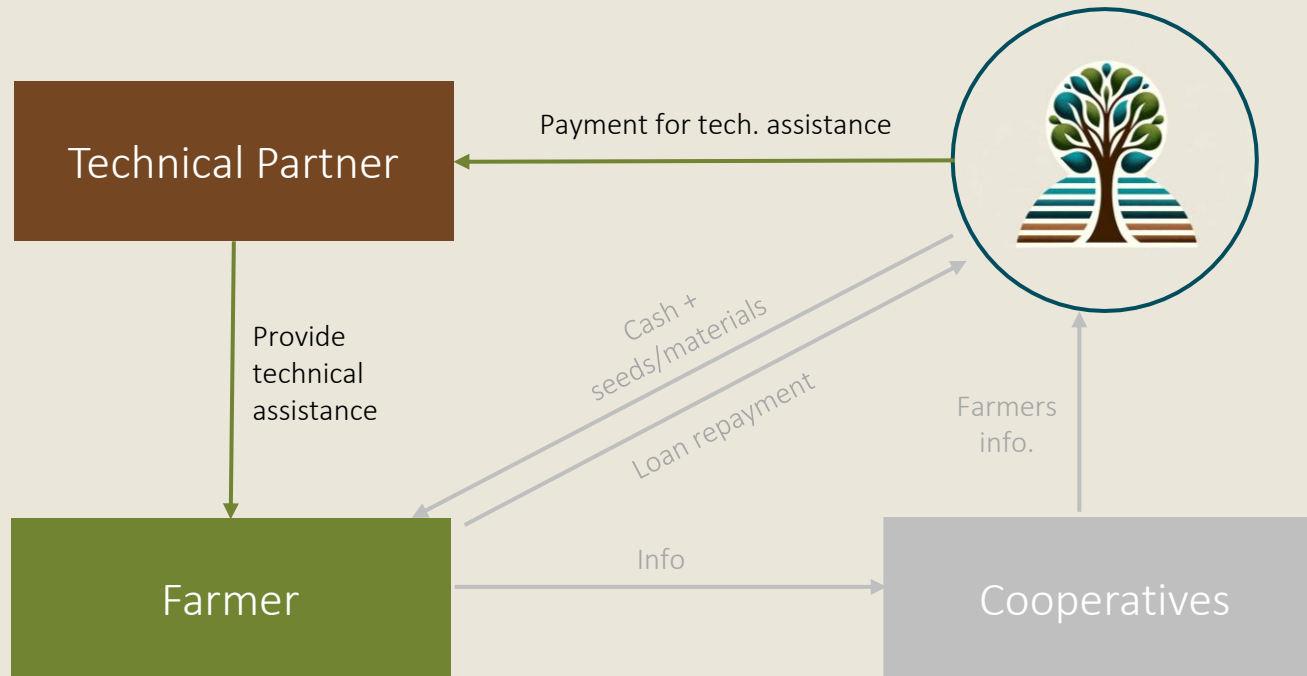


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Caatinga Bank will partner with local companies to provide technical support to farmers



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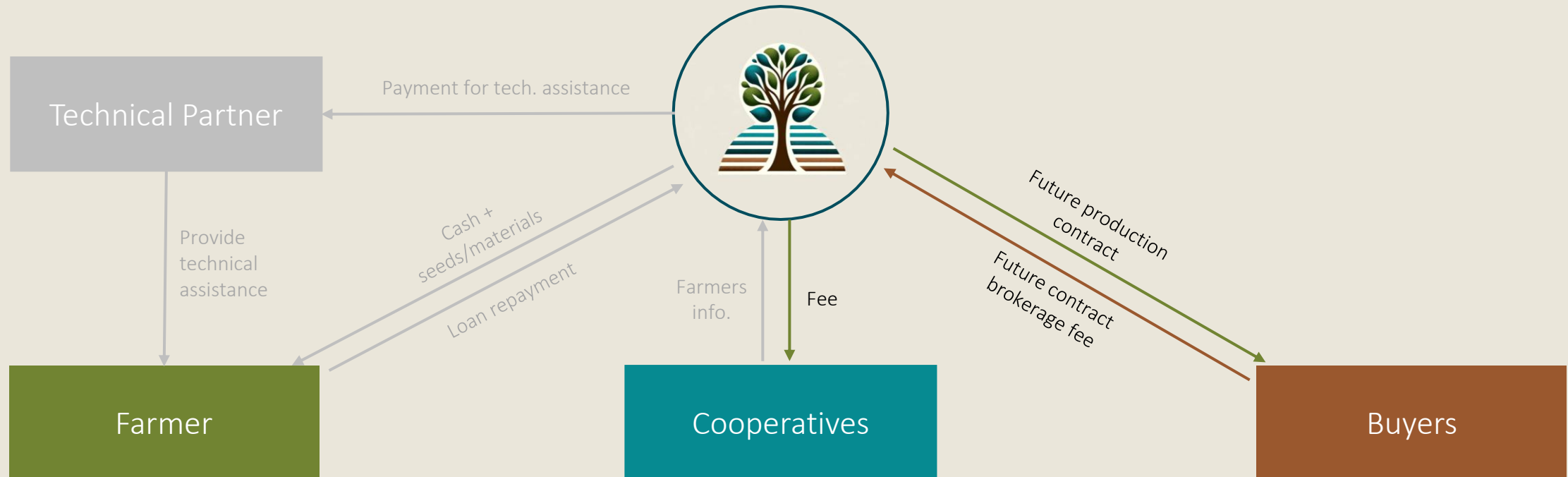
Sustainable production practices:

1. Intercropping, crop succession or crop rotation
2. Climate-smart crops adoption
3. Soil quality analysis

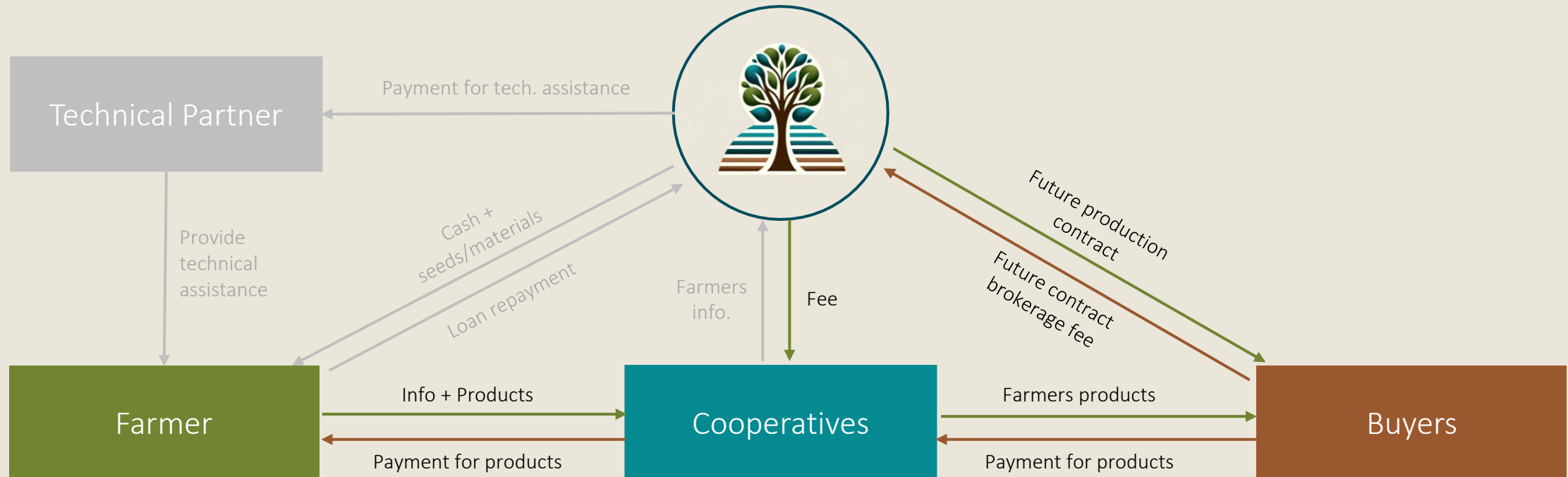
To diminish the risk of default, Caatinga Bank will secure future contracts with regional buyers



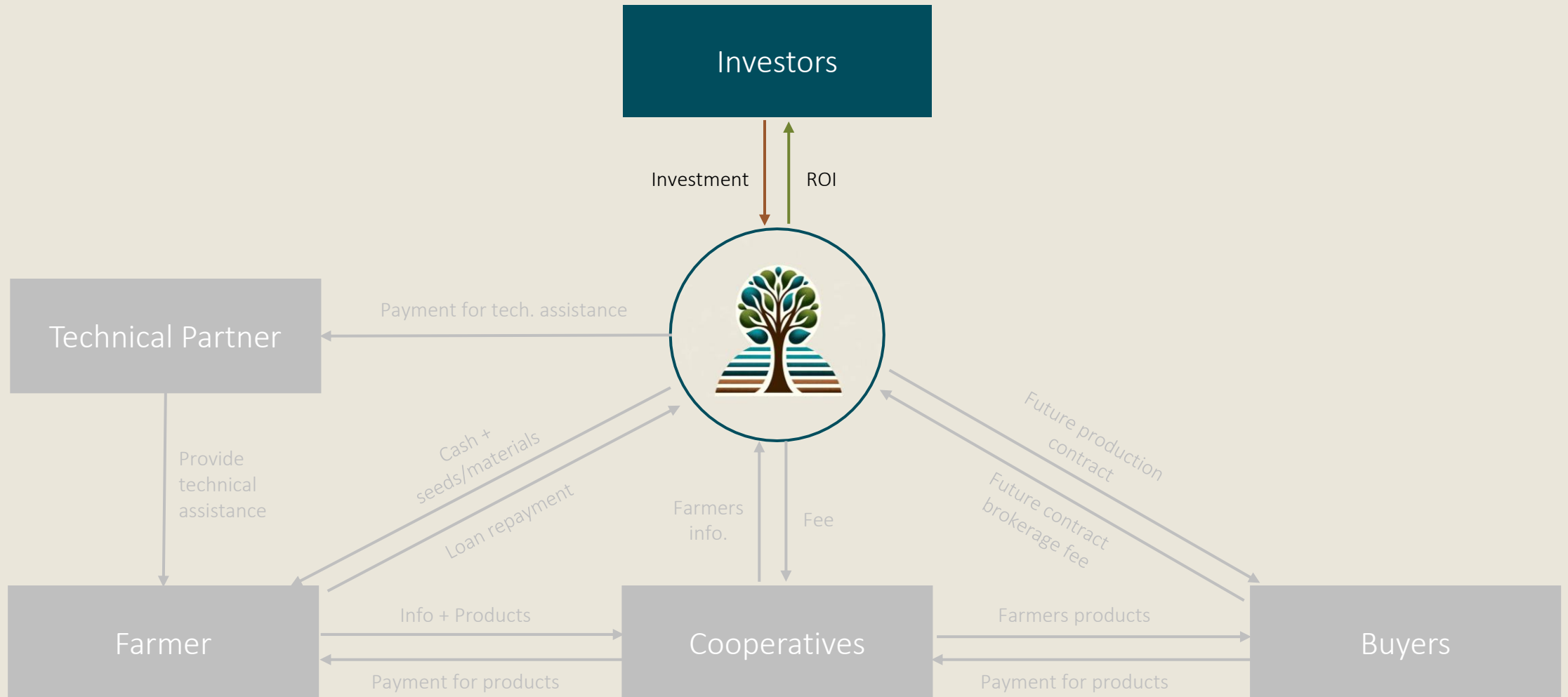
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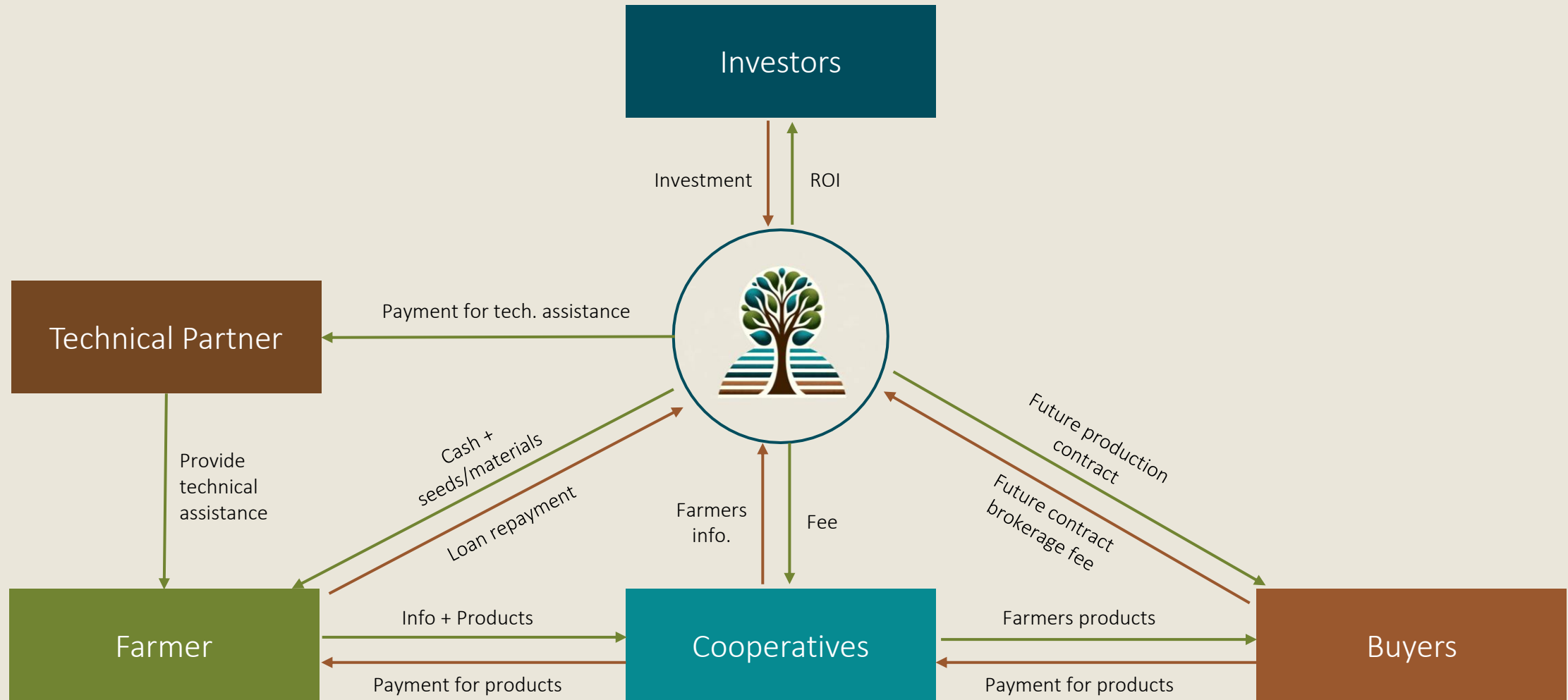
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A blended investment model made of private and governmental investors will fund the Caatinga Bank operation



CB model accounts for several stakeholders to guarantee operation and increase credibility of our bank



Partnerships with existing semiarid organizations are essential to initiate the Caatinga Bank implementation

Farmer

Pilot program in the cities of Petrolina and Juazeiro (1,000 farmers)

Expand to the whole region (target: +68.2k farmers)



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Investors

Blended finance - support from national programs for initial investment and attracting private investment



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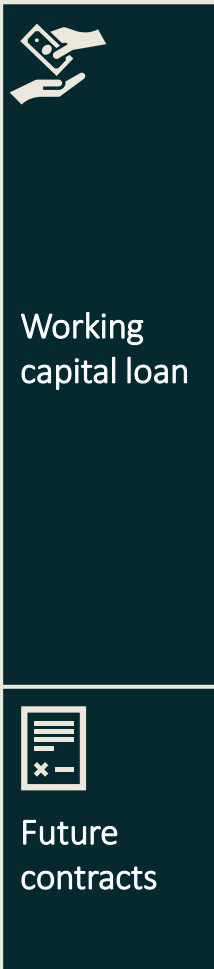
Buyers

Partnership with the main buyers of the region

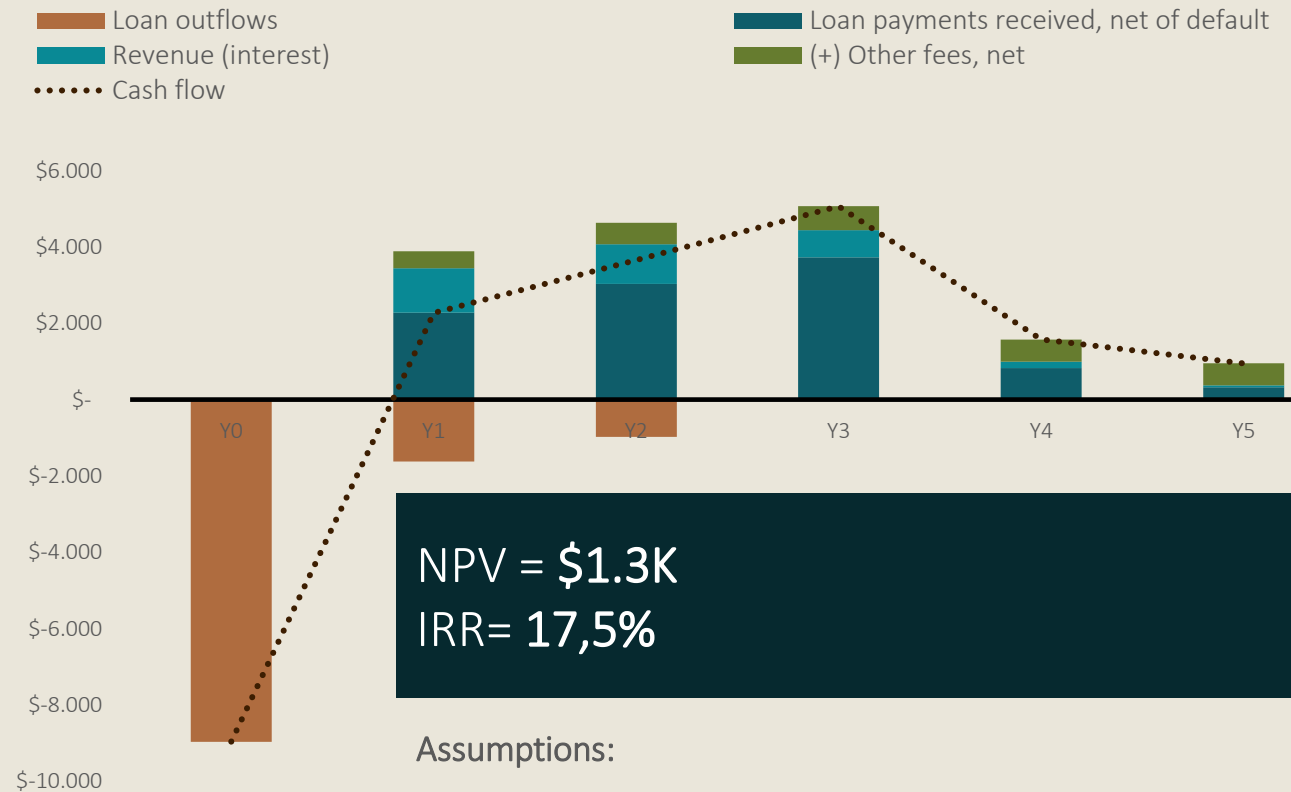


Average lifecycle of the initiative for 1 farmer is 5 years and its unit economics shows positive NPV and IRR higher than risk free rate

Financial products details:



Unit Economics – Cash flow (USD)



Assumptions:

- Default rate of 9,2% per year (*conservative benchmark aligned with sector and geography*)
- Discount rate of 11,25% (*SELIC – Risk free rate*)

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Working capital loan

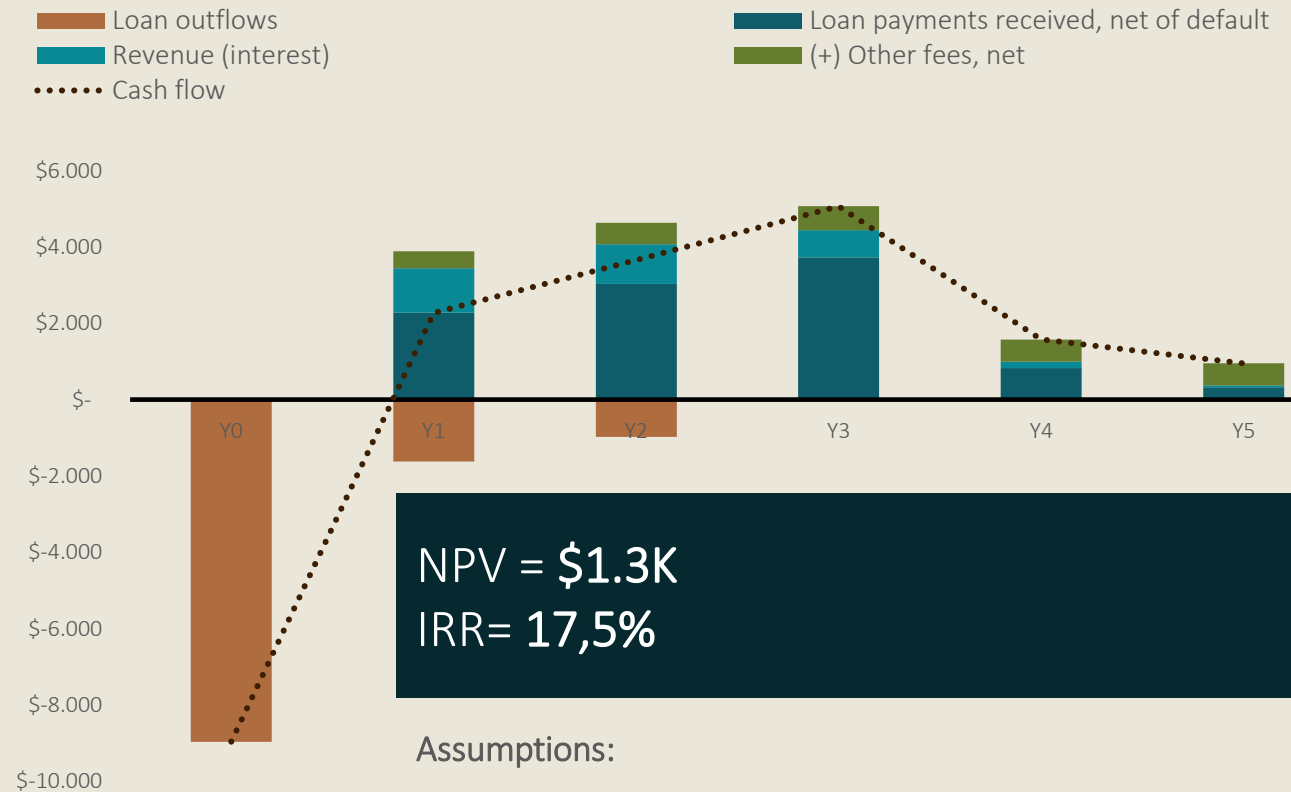


Future contracts

Financing:

- \$11.2k per farmer loan outflows present value

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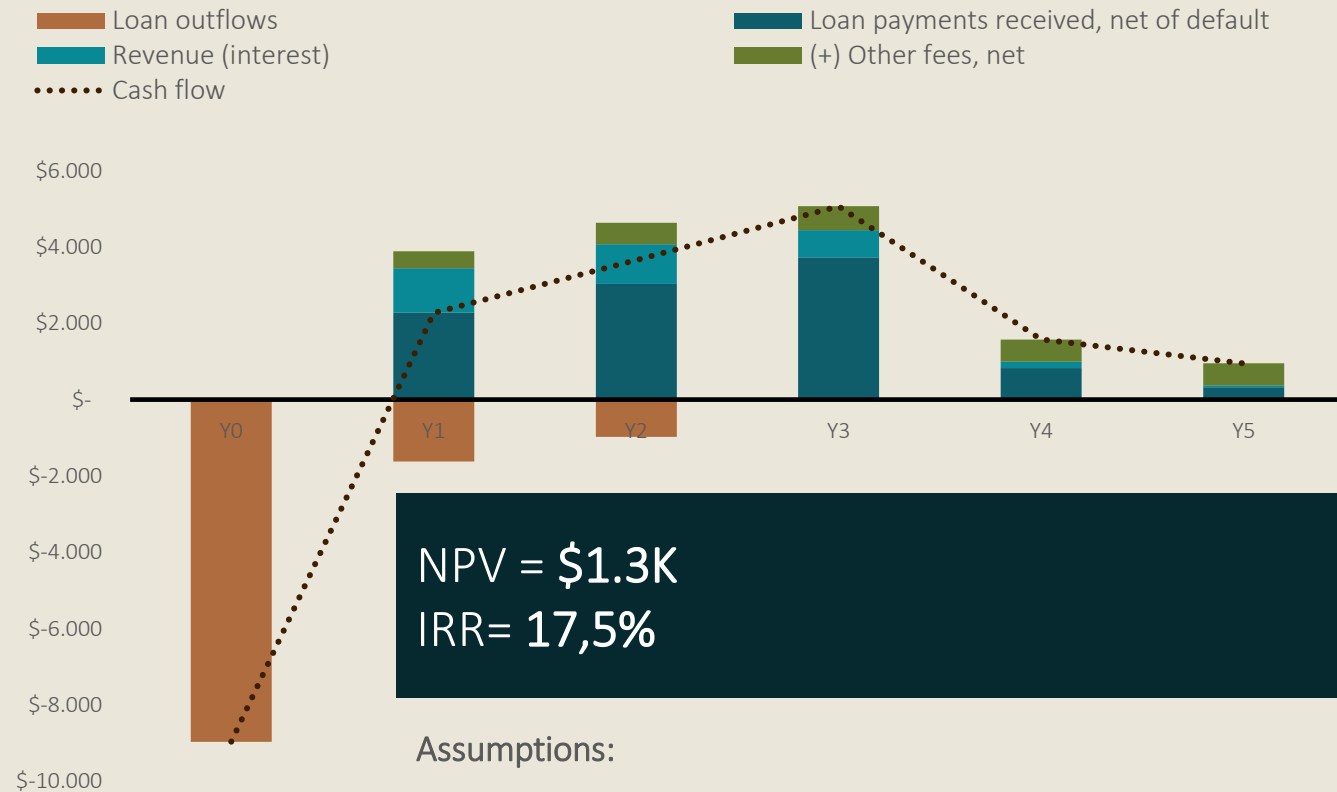
Repayment loan terms:

- 3 years payment terms, last payment on Y5



Future contracts

Unit Economics – Cash flow (USD)



Assumptions:

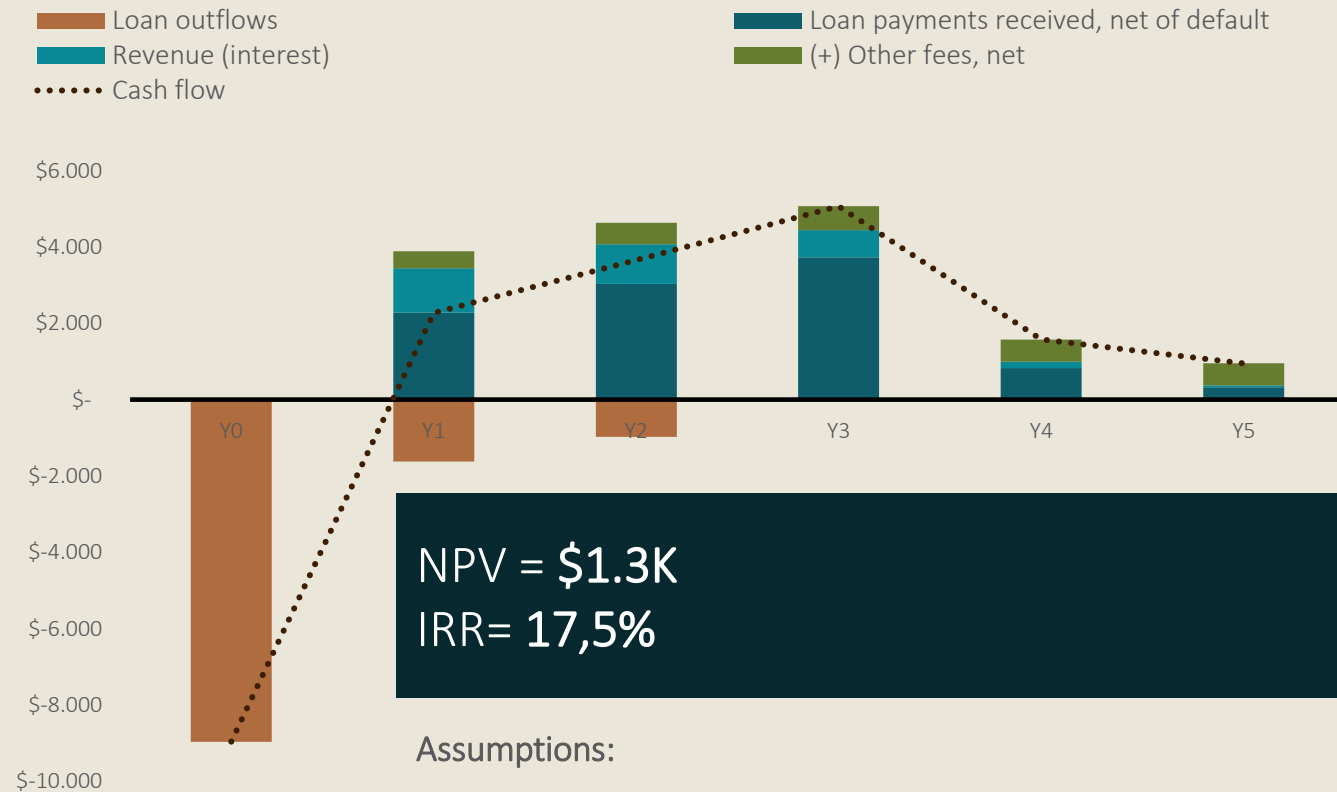
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Financial products details:

 Working capital loan	Financing: \$11.2k per farmer loan outflows present value
	Repayment loan terms: 3 years payment terms, last payment on Y5
	Interest terms: 13% p.a. Interest rate, aligned with market benchmarks
 Future contracts	

Unit Economics – Cash flow (USD)



Assumptions:

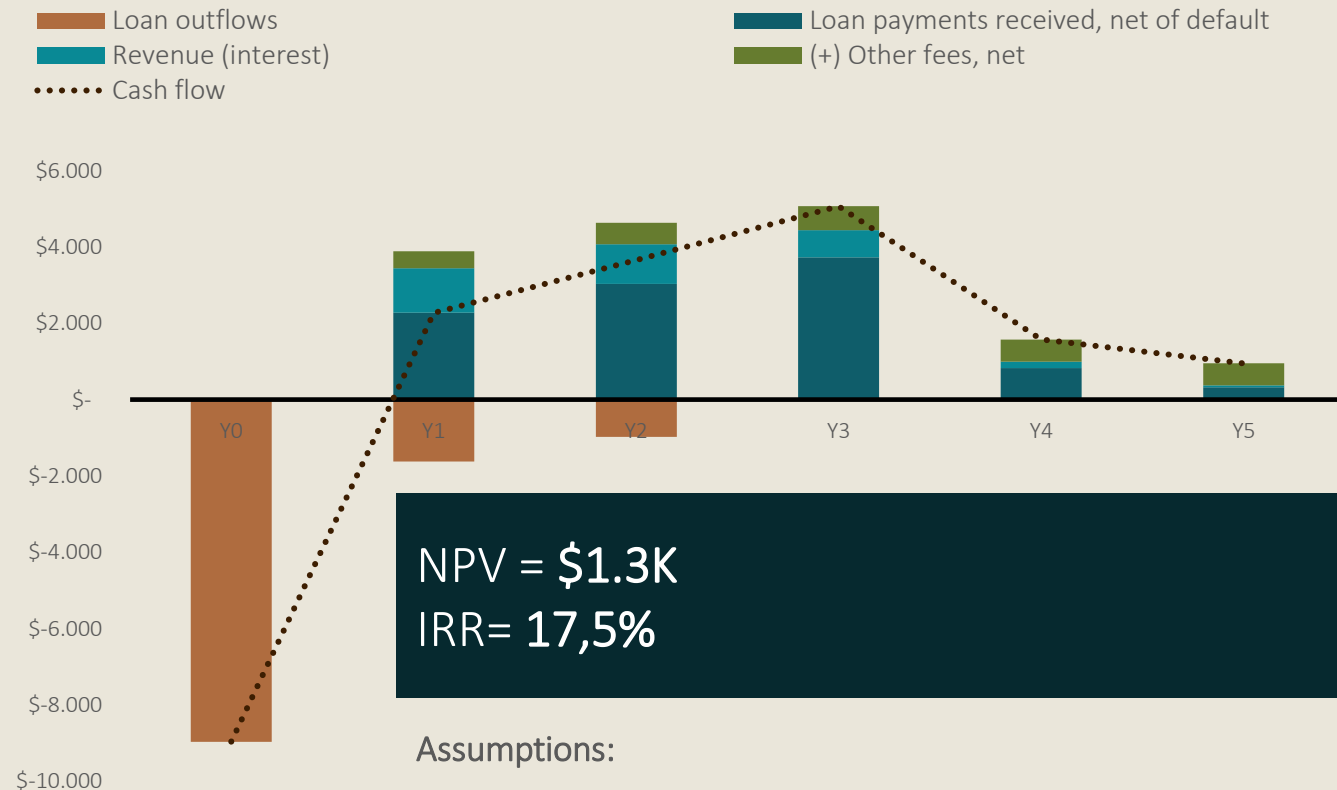
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Unit Economics – Cash flow (USD)

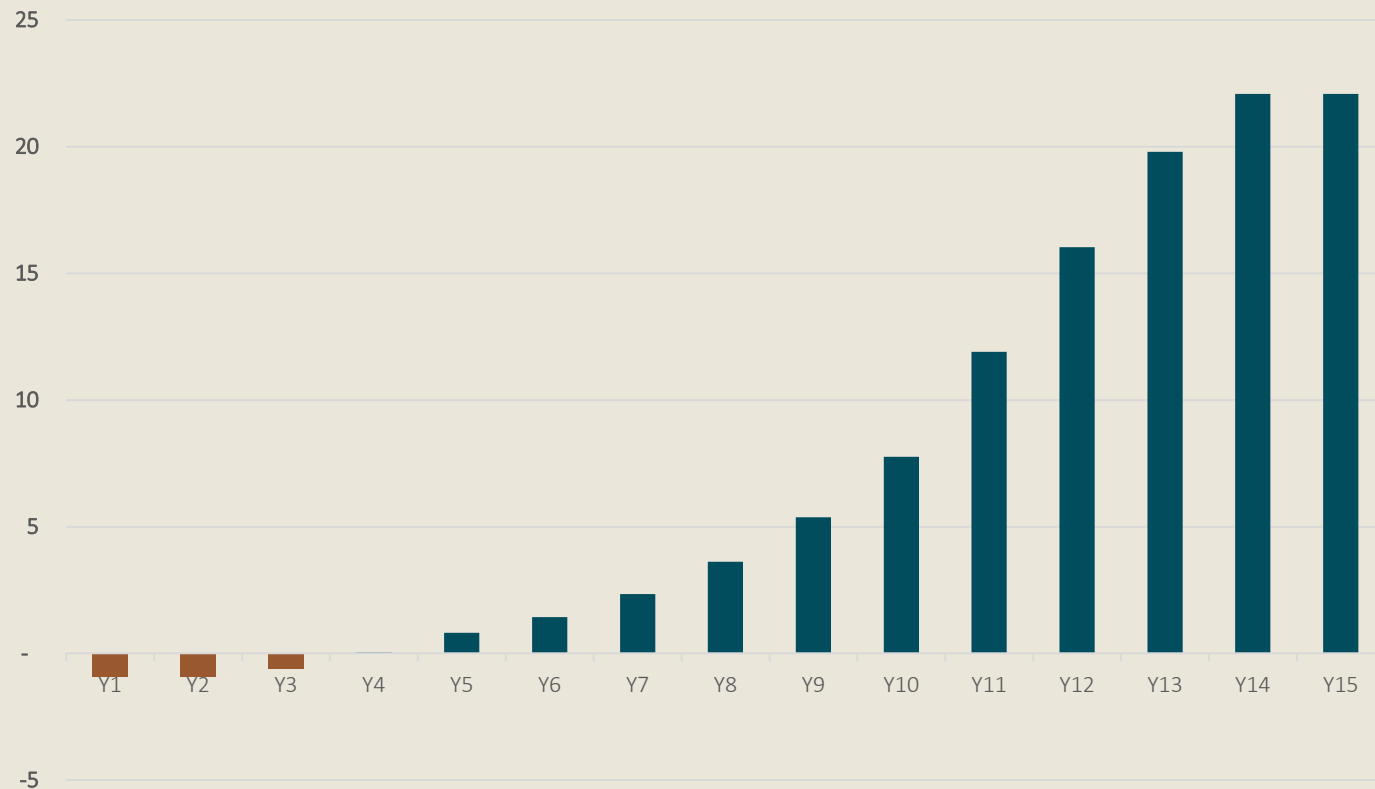


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Scaling up, Caatinga Bank achieves breakeven on Y4 and at steady state achieves a profit of ~22M USD and ROS of ~16%

Caatinga bank projected net profits – Year on Year USD (M)



Financial details:

Investment size	 Target: +68.2k Farmers (5% of the addressable market) Approx. \$12M fund on 1 st year
Revenue sources	 Interest rate: 13% per year Brokerage fee: 5% over future agricultural contract
Operating costs	 Yield paid to Caatinga Bank's investors: 14% - 1,75%. higher than risk free interest rate <i>In line with expected returns of main banks in Brazil</i>
	Fees paid to cooperatives: 20% of future contract's brokerage fee
	Cost of technical partner: \$40/farmer on first year and \$10/farmer in years 2 to 5
	Other operating costs: Salaries, office administration, legal and compliance fees, taxes

A mitigation plan is in place to address risks related to banking activities and the geographic operations

Banking activities

<u>Risks</u>	<u>Mitigation</u>
Credit risk: Farmers not repaying the loan	• Cooperative fee payments linked to default rates of their farmers • Conservative default rate on financial model • Future contracts
Liquidity risk: Lack of cash to meet obligations	• Diversified funding sources – blended financing with public & private sources of funding • Cash flow forecasting & control
Fraud risk: Farmers not using the credit for farming purposes	• Cooperatives making regular visits to farmers; tying credit to sources of investment other than cash (e.g. seeds) • Cooperatives have shared accountability for referrals

Geographic location

<u>Risks</u>	<u>Mitigation</u>
Market risk: Farmers not selling crops; adverse effects of market – e.g. price changes, crisis	• Future contracts • Diversified investments • Diversify location and crop types (within caatinga region)
Regulatory risk: Not complying with regulatory standards for banks in Brazil	• Keep 16% Basel level, 2x the legal requirement for banks in Brazil • Business policies + monitor & track regulatory policies and its changes
Operational risk: Difficulty to reach out partners; low adherence on technical program	• Work with cooperatives to gain access to farmers' network (+fee to incentivize) • Loan covenant to ensure participation in the technical support program



Caatinga Bank will be one of the agents to help ending the cycle of poverty in the Caatinga region



+50% increase in farms productivity



22% of degraded land recovered



15x increase in farmers profit



16% Caatinga Bank net margin

WHAT SDG Goals	WHO Stakeholders impacted
 [1.4] Economic resources: increase access to credit and professionalizing services.	Small farmers in the Brazilian Caatinga <i>KPI: % of small farmers in Caatinga who have access to credit and professional support</i>
 [8.3] Growth of SMEs: increase productive activities and formalization	Small farmers in the Brazilian Caatinga <i>KPI: economic growth in the Caatinga region</i>
 [10.2] Social inclusion: promote economic and social inclusion	Small farmers in the Brazilian Caatinga, especially those who today live through subsistence agriculture <i>KPI: employment among population in Caatinga</i>
 [12.2] Use of natural resources: increase efficient use of natural resources	Caatinga Biome <i>KPI: carbon emissions in Caatinga</i>
 [15.3] Combat desertification: restore degraded land	Caatinga Biome <i>KPI: acres of land restored</i>

Team



Carolina Pascotto

*MBA Associate in
search fund:*



*Product Marketing manager
at Brazilian's biggest fintech*



Danielle van Drunen

*Risk manager in
LATAM Fintech:*



Consultant in:



Gabriela Galera

*Market insights lead
in:*



Client Manager in:



Luana Nogueira

*Operations Manager
in Brazilian NGO:*



Consultant in:





Thank you!

Appendix

Financial Model – Farmer's cashflow

P&L and Cashflow Farmer - USD

	Baseline	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Annual revenue	9.289	11.147	12.540	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933
<i>Improvement in operations</i>	0	4%	7%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
Annual costs	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968
Farmer's Profit	321	2.178	3.572	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965
Return on sales		24%	40%	55%	55%	55%	55%	55%	55%	55%	55%	55%	55%	55%	55%	55%
Interest paid to banks		1.166	1.034	713	169	48	-	-	-	-	-	-	-	-	-	-
Final Cash to Farmer	1.013	2.538	4.252	4.796	4.918	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965	4.965
Increase in Farmer's profit		3	8	13	15	15	15	15	15	15	15	15	15	15	15	15
					4.796,48											
Cash saved - %		0%	0%	0%	30%	40%	50%	70%	70%	70%	70%	70%	70%	70%	70%	70%
Cash saved for farmer's life	-	-	-	-	1.439	1.967	2.483	3.476	3.476	3.476	3.476	3.476	3.476	3.476	3.476	3.476
Cash Reinvested in operations	1.013	2.538	4.252	3.358	2.951	2.483	1.490	1.490	1.490	1.490	1.490	1.490	1.490	1.490	1.490	1.490
Cash at beginning of period	0	7.348	7.991	8.970	11.376	13.941	16.423	17.913	19.402	20.892	22.381	23.871	25.361	26.850	28.340	29.829
Credit Needed	8.968	1.620	977	-	-	-	-	-	-	-	-	-	-	-	-	-
Cost of production	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968	8.968
Sales generated	11.147	12.540	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933	13.933
Payment Principal	2.632	3.450	4.185	931	366	-	-	-	-	-	-	-	-	-	-	-
Interest Paid	1.166	1.034	713	169	48	-	-	-	-	-	-	-	-	-	-	-
Fee Paid	0	65	65	20	20	0	0	0	0	0	0	0	0	0	0	0
Cash saved for farmer's life	-	-	-	1.439	1.967	2.483	3.476	3.476	3.476	3.476	3.476	3.476	3.476	3.476	3.476	3.476
Cash at end of the period	7.348	7.991	8.970	11.376	13.941	16.423	17.913	19.402	20.892	22.381	23.871	25.361	26.850	28.340	29.829	29.829

Financial Model – Unit Economics detail

Loan unit economics (1 farmer) - USD

		0	1	2	3	4	5
Loan given	\$	-8.968	\$ -1.620	\$ -977	\$ -	\$ -	\$ -
Repayments Principal		\$	2.632	\$ 3.450	\$ 4.185	\$ 931	366
Interest Payments		\$	1.166	\$ 1.034	\$ 713	\$ 169	48
(-) Default		\$	-349	\$ -413	\$ -451	\$ -101	-38
(+) Other fees, net		\$	446	\$ 567	\$ 622	\$ 577	577
Subtotal		\$	3.895	\$ 4.638	\$ 5.070	\$ 1.576	953
Cash flow	\$	-8.968	\$ 2.275	\$ 3.661	\$ 5.070	\$ 1.576	953

Loan given - Present value	\$ 11.213,90
NPV - Cash flow	\$ 1.304,94
IRR	17,5%
Risk free rate	11,25%

Financial Model – Unit Economics detail

Bank - Total projections (USD)

	1	2	3	4	5	6	7	8	9	10	Run-rate				
											11	12	13	14	15
Credit Granted to farmers	9.622.460	15.026.829	21.801.233	30.108.197	41.580.380	57.423.830	79.304.139	109.521.543	151.252.741	208.884.855	221.475.728	226.763.146	226.763.146	226.763.146	226.763.146
Credit Received from farmers	2.824.395	7.602.258	14.989.676	21.700.417	30.362.020	41.930.917	57.907.933	79.972.700	110.444.848	152.527.854	190.979.649	218.308.117	224.780.883	226.762.598	226.762.598
Cumulative credit	9.622.460	21.824.894	36.023.869	51.142.390	71.022.353	98.084.163	135.457.384	187.070.995	258.351.035	356.791.043	425.738.916	461.522.414	469.977.443	471.959.706	471.960.254
# New farmers	1.073	1.482	2.046	2.826	3.903	5.390	7.444	10.280	14.197	19.607	19.607	19.607	19.607	19.607	19.607
# Old farmers	-	1.073	2.555	4.601	6.354	8.775	12.119	16.737	23.114	31.922	44.085	53.412	58.821	58.821	58.821
# Farmers churn (dont need us any	-	-	-	1.073	1.482	2.046	2.826	3.903	5.390	7.444	10.280	14.197	19.607	19.607	19.607
# Total # of Farms	1.073	2.555	4.601	6.354	8.775	12.119	16.737	23.114	31.922	44.085	53.412	58.821	58.821	58.821	58.821
Revenue	2.132.022	5.013.413	8.570.243	12.867.543	18.669.452	25.783.108	35.607.294	49.174.808	67.911.978	93.788.605	114.679.843	129.124.765	137.608.712	142.141.171	142.141.171
Contribution Margin (Total)	1.541.846	3.625.625	6.197.871	9.305.613	13.501.466	18.645.955	25.750.658	35.562.479	49.112.917	67.826.504	82.934.732	93.381.082	99.516.544	102.794.350	102.794.350
Fixed costs															
Yield to investors	1.347.144	3.055.485	5.043.342	7.159.935	9.943.129	13.731.783	18.964.034	26.189.939	36.169.145	49.950.746	59.603.448	64.613.138	65.796.842	66.074.359	66.074.436
Salaries	715.008	752.248	807.576	891.100	966.644	1.077.832	1.170.400	1.262.968	1.401.820	1.662.500	1.662.500	1.662.500	1.662.500	1.662.500	1.662.500
Office administration	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520	11.520
Legal compliance fees	77.092	181.281	309.894	465.281	675.073	932.298	1.287.533	1.778.124	2.455.646	3.391.325	3.391.325	3.391.325	3.391.325	3.391.325	3.391.325
Other general and administrative ex	200.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000	400.000
EMBRAPA - Project developm	107.295	148.178	204.638	282.612	390.296	539.011	744.392	1.028.029	1.419.741	1.960.707	1.960.707	1.960.707	1.960.707	1.960.707	1.960.707
EMBRAPA - Technical suppo	-	10.730	25.547	35.282	48.725	67.291	92.931	128.340	177.242	244.777	244.777	244.777	244.777	244.777	244.777
Total Fixed costs	2.458.060	4.559.442	6.802.517	9.245.729	12.435.388	16.759.735	22.670.809	30.798.921	42.035.114	57.621.576	67.274.278	72.283.968	73.467.672	73.745.188	73.745.265
Subtotal	- 916.213	- 933.817	- 604.646	- 59.884	- 1.066.078	- 1.886.220	- 3.079.849	- 4.763.559	- 7.077.804	- 10.204.928	- 15.660.454	- 21.097.114	- 26.048.872	- 29.049.162	- 29.049.085
Taxes	-	-	-	14.372	255.859	452.693	739.164	1.143.254	1.698.673	2.449.183	3.758.509	5.063.307	6.251.729	6.971.799	6.971.780
Profit	- 916.213	- 933.817	- 604.646	45.512	810.219	1.433.527	2.340.685	3.620.305	5.379.131	7.755.746	11.901.945	16.033.807	19.797.143	22.077.363	22.077.305
RDS	-43%	-19%	-7%	0%	4%	6%	7%	7%	8%	8%	10%	12%	14%	16%	16%

*The Reverte Program

The Reverte Program was created to support rural producers in recovering degraded areas through an integrated solution that involves good **agricultural practices, financial tools, and input-use protocols** that make fertilizers and seeds to machinery and pesticides appropriate for growing soy and other intercropped harvests. Through **partnerships** that share the same objectives, the Program also seeks to strengthen pre-existing projects that can contribute to the evolution and **scale of regenerative agriculture in the Cerrado**.

In the first phase of the Program and the credit line, eight producers benefited from the funds, which are being invested in 19 farms, covering a total area of **144,600 ha**, of which 55,300 ha are native vegetation according to self-declared CAR information on the rural properties included in the Program. In these farms, **31.400ha of degraded pastureland will be recovered using resources from the line of credit**.

Link to article:

<https://www.nature.org/en-us/about-us/where-we-work/latin-america/brazil/stories-in-brazil/restoring-degraded-landscapes-in-the-cerrado/#:~:text=The%20Reverte%20Program%20was%20created,growing%20soy%20and%20other%20intercropped>